

Experiment 2: Diodes and Applications

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EEE3307-C0013: Electronics I

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2.A Objective

To study the characteristics and the applications of PN junction diodes

2.B Pre-lab

2.B.1 Rectifiers

The first circuit we set up was a simple half-wave rectifier. We first simulated the circuit in LTSpice. We chose $100\mu F$ and $2k\Omega$. Figure 1 shows the circuit without the capacitor and Figure 2 shows the circuit with the Resistor. Figure 3 shows the circuit with both in.

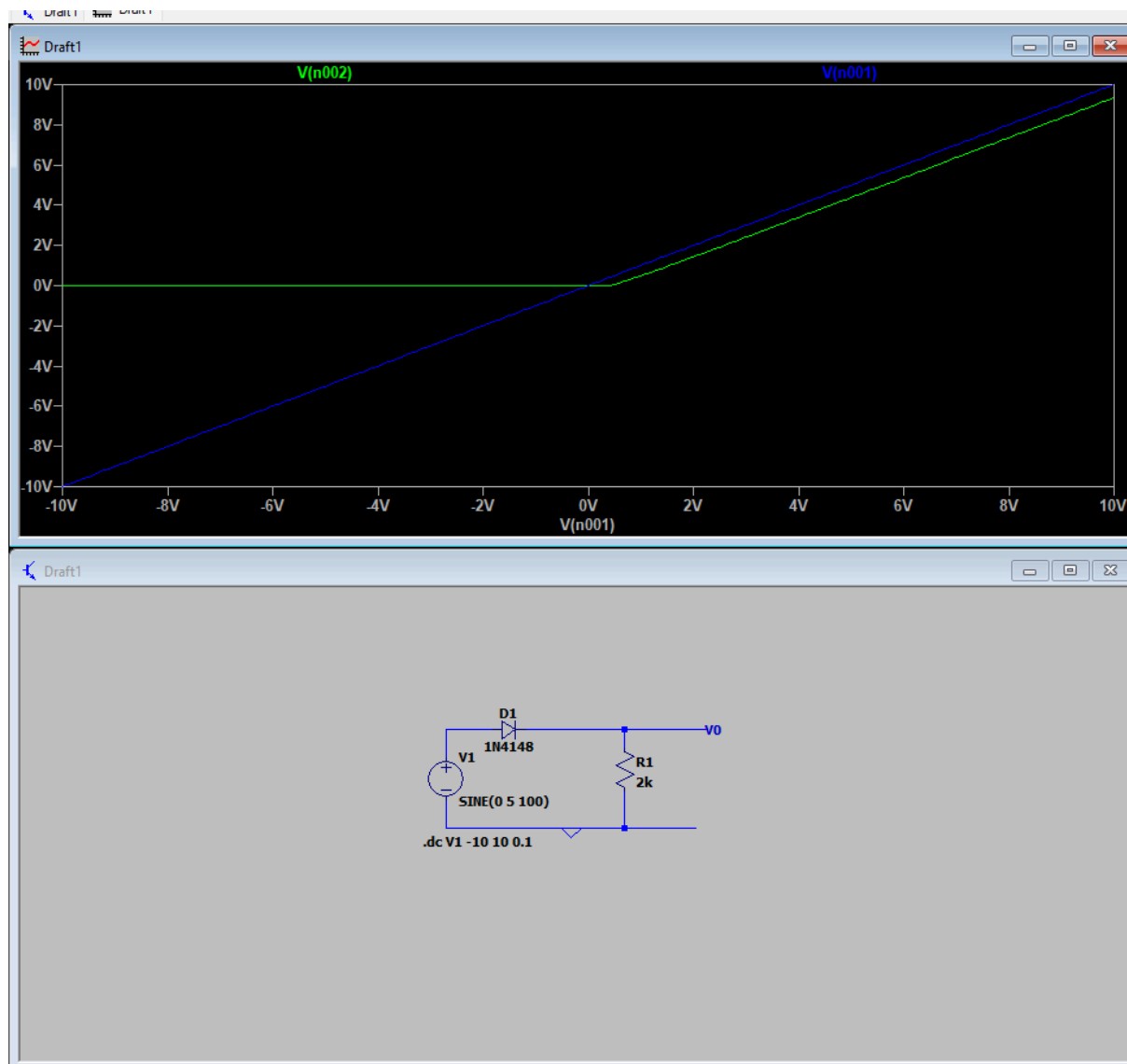


Figure 1: “Half-wave rectifier without capacitor”

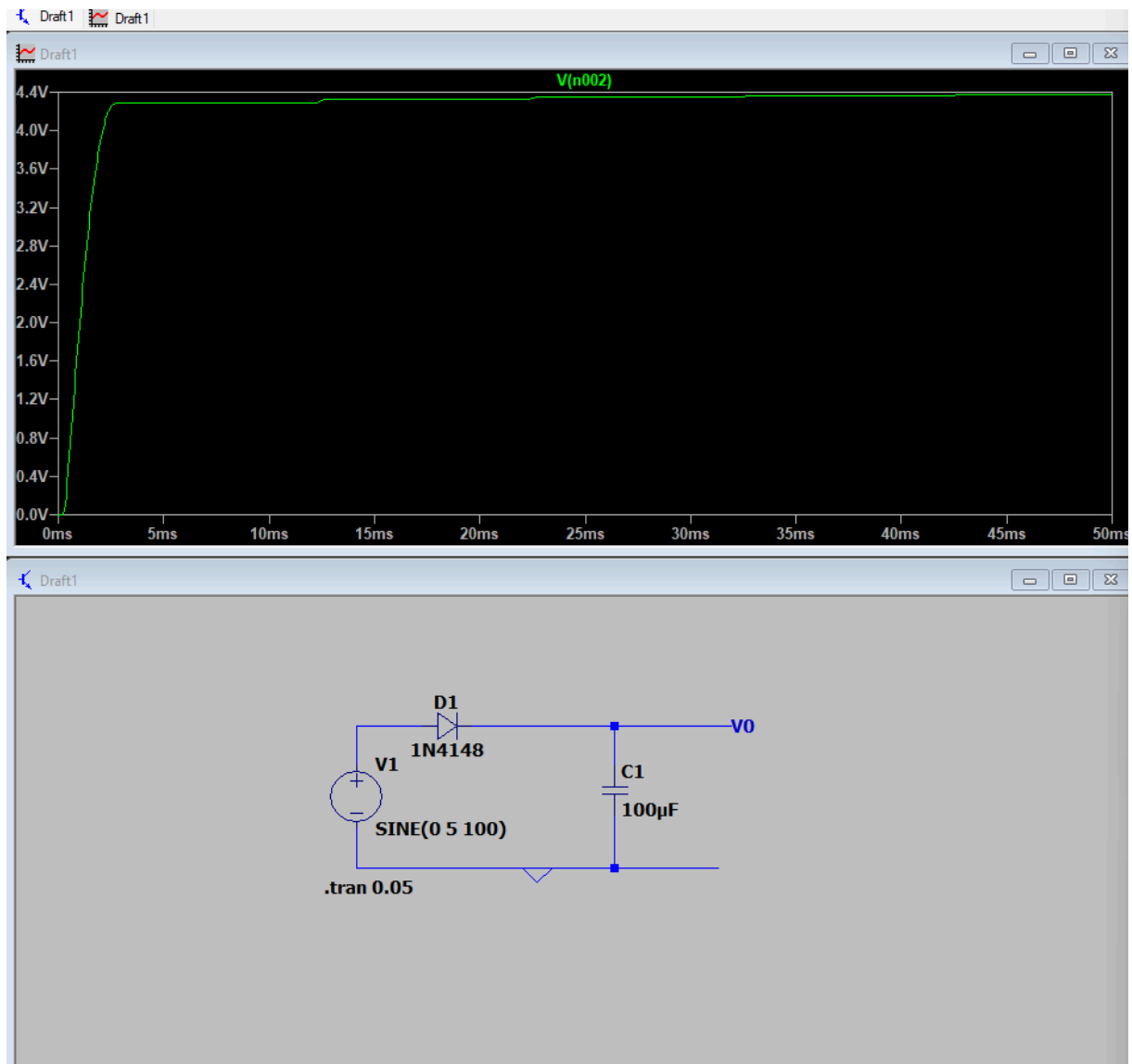


Figure 2: “Half-wave rectifier without resistor”

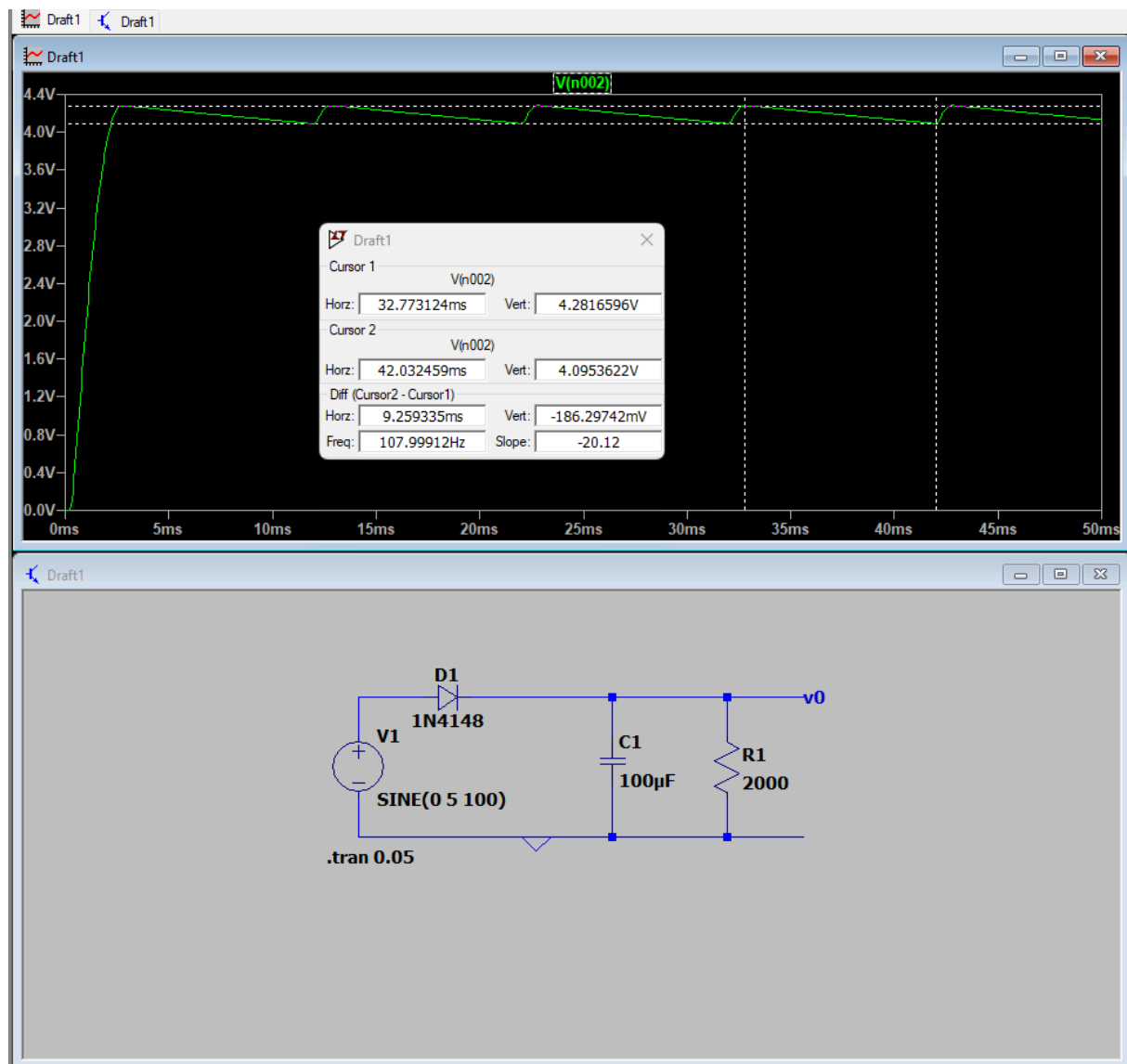


Figure 3: “Half-wave rectifier with both capacitor and resistor”

The next circuit was a full-wave rectifier. We first simulated the circuit in LTSpice. We chose $100\mu F$ and $2k\Omega$. Figure 4 shows the circuit without the capacitor and Figure 5 shows the circuit with the Resistor. Figure 6 shows the circuit with both in.

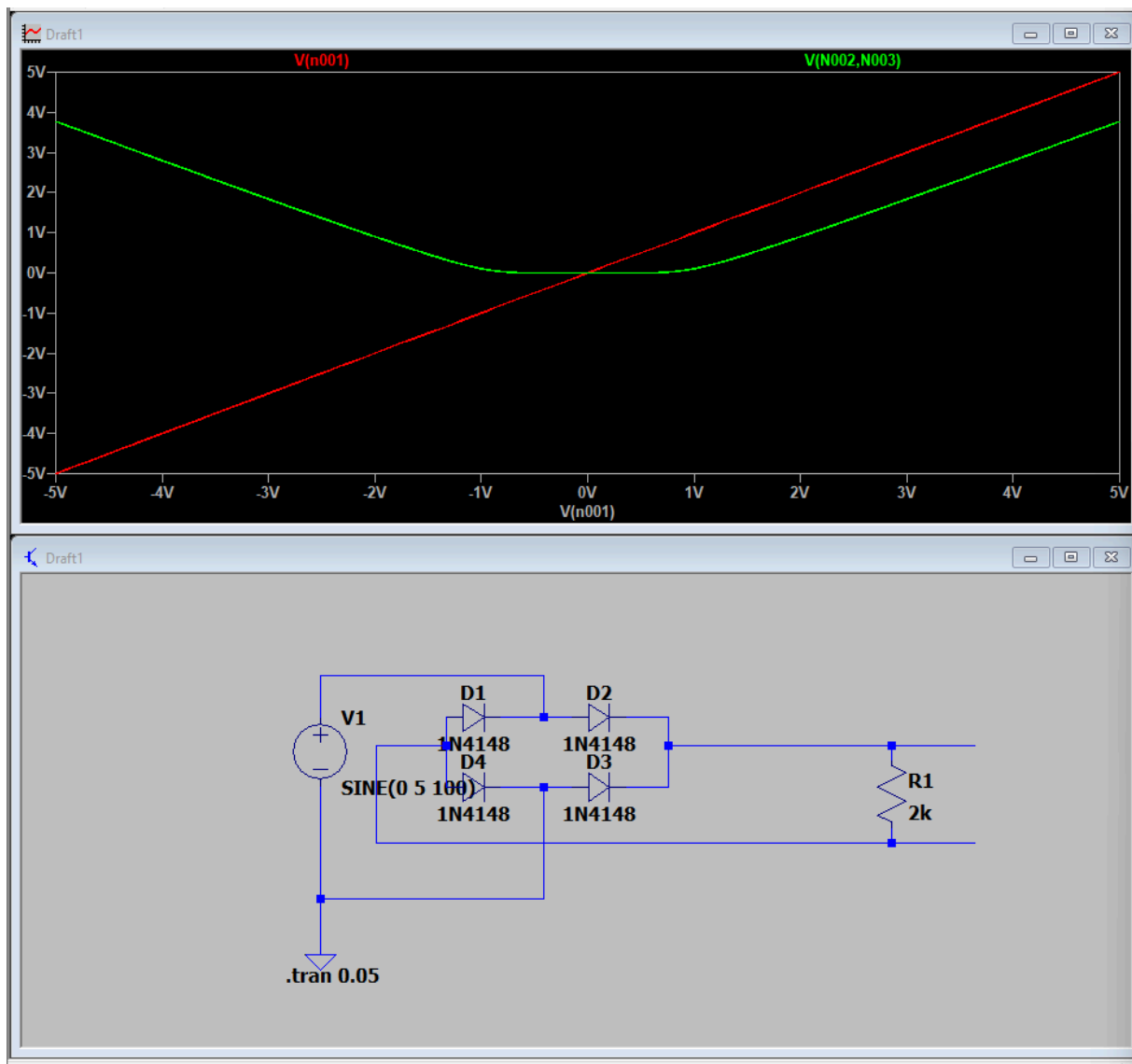


Figure 4: “Half-wave rectifier without capacitor”

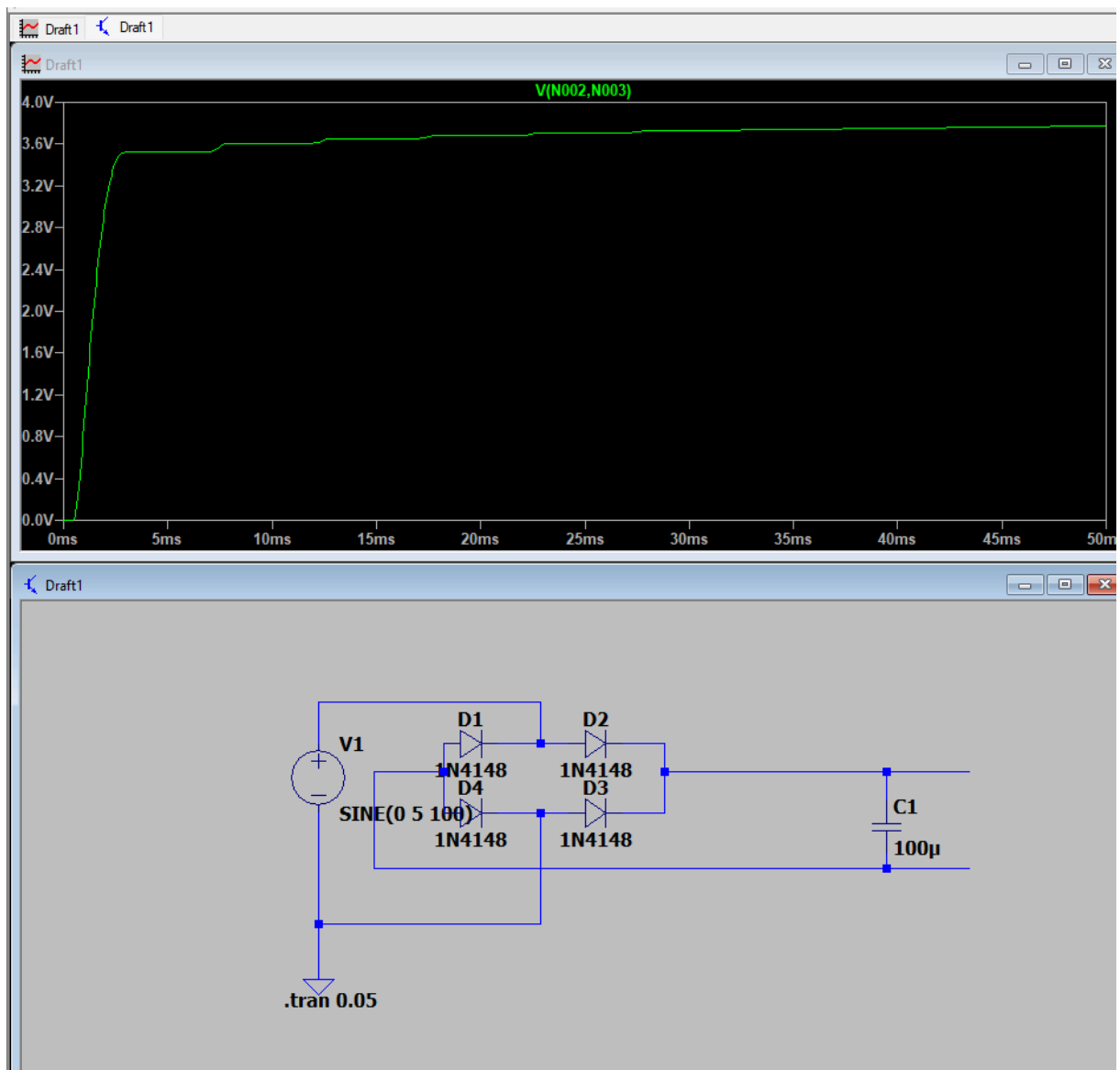


Figure 5: “Half-wave rectifier without resistor”

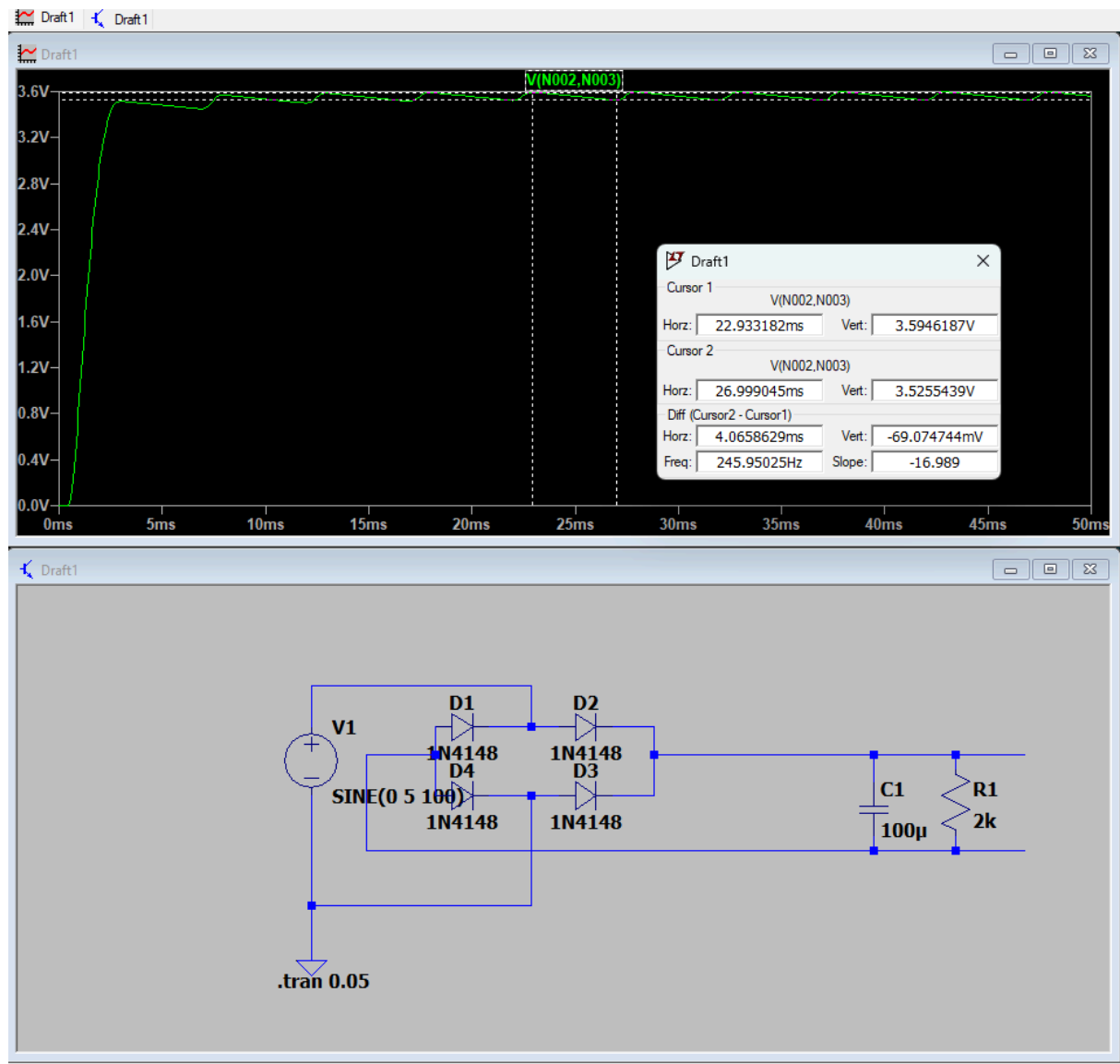


Figure 6: “Half-wave rectifier with both capacitor and resistor”

2.B.2 Clipping circuits

We determined the transfer characteristics and output waveforms of the clipper circuits using LTSpice. The first circuits transfer characteristics are showing in Figure 7 and the output waveforms are shown in Figure 8. The second circuits transfer characteristics are showing in Figure 9 and the output waveforms are shown in Figure 10.

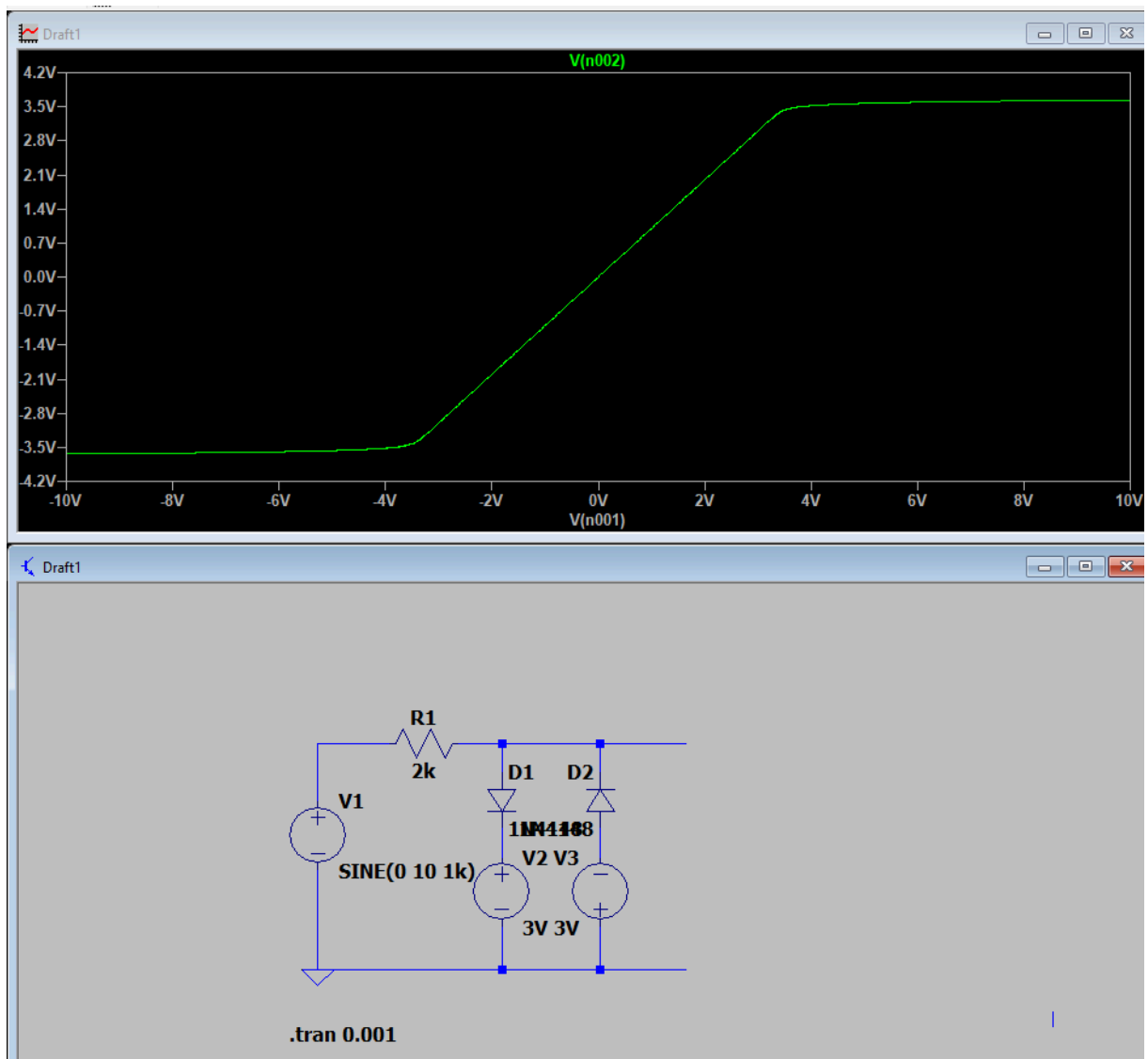


Figure 7: “Clipping circuit 1 transfer characteristics”

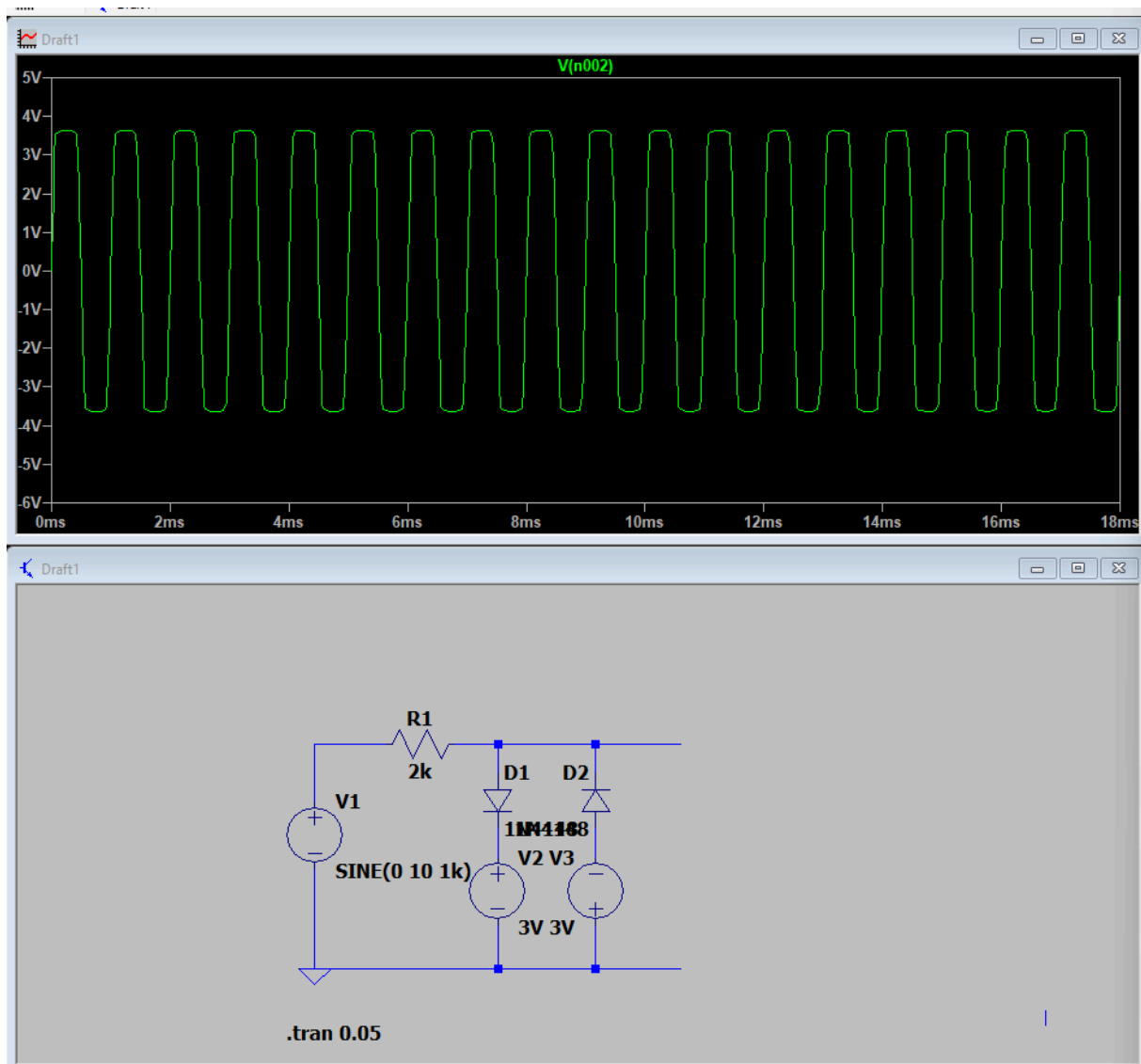


Figure 8: “Clipping circuit 1 output waveforms”

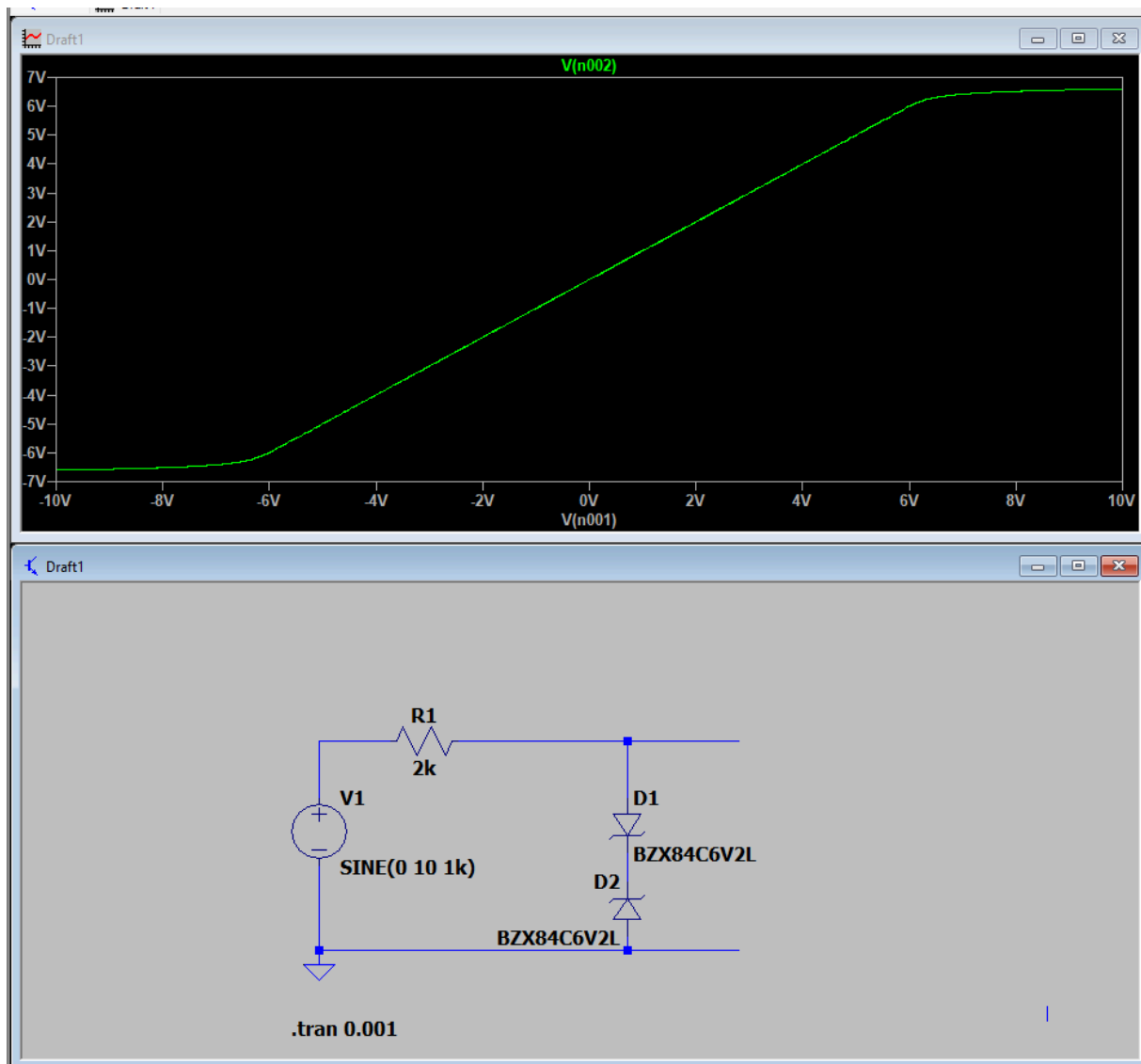


Figure 9: "Clipping circuit 2 transfer characteristics"

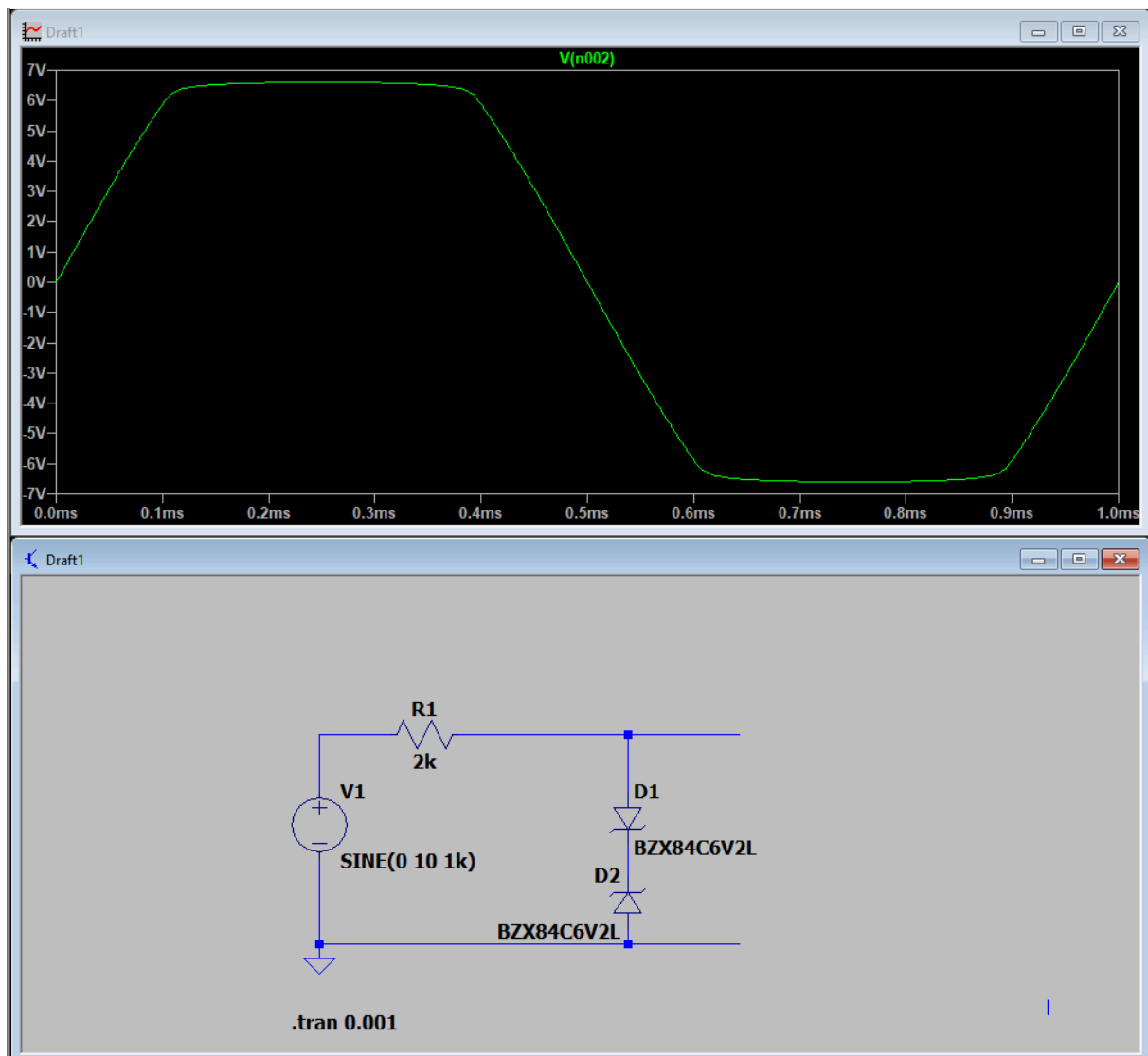


Figure 10: “Clipping circuit 2 output waveforms”

2.B.3 Clamping circuits

For the clamping circuit, we used the values in Equation 1. The output is shown in Figure 11.

$$\begin{aligned}
 R &= 2k\Omega \\
 C &= 100\mu F \\
 V_B &= 3V \\
 \text{Diode: } &1N4148
 \end{aligned}
 \tag{Eq 1}$$

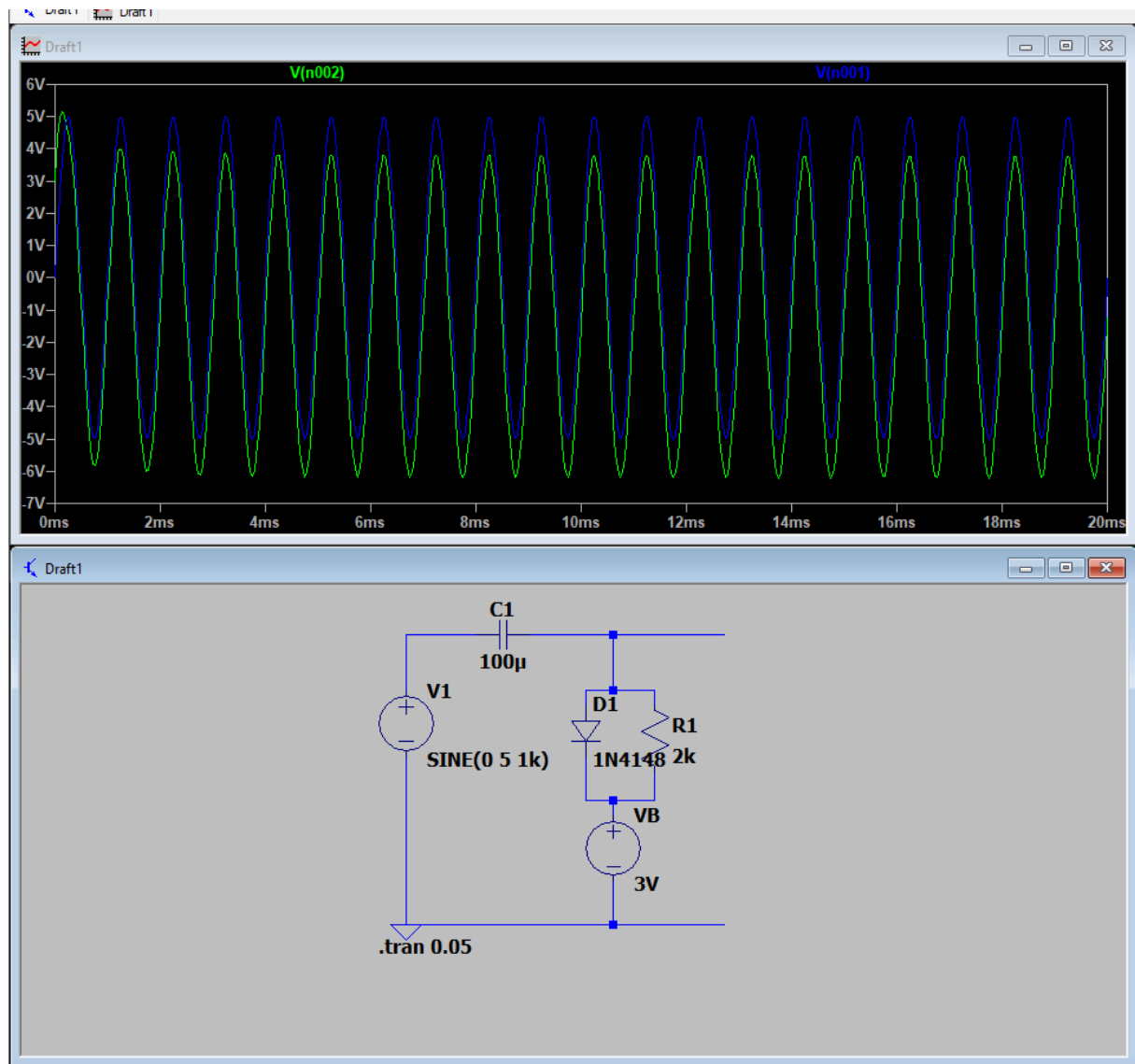


Figure 11: “Clamping circuit output waveforms”

2.C Experiment

Experimental results are below

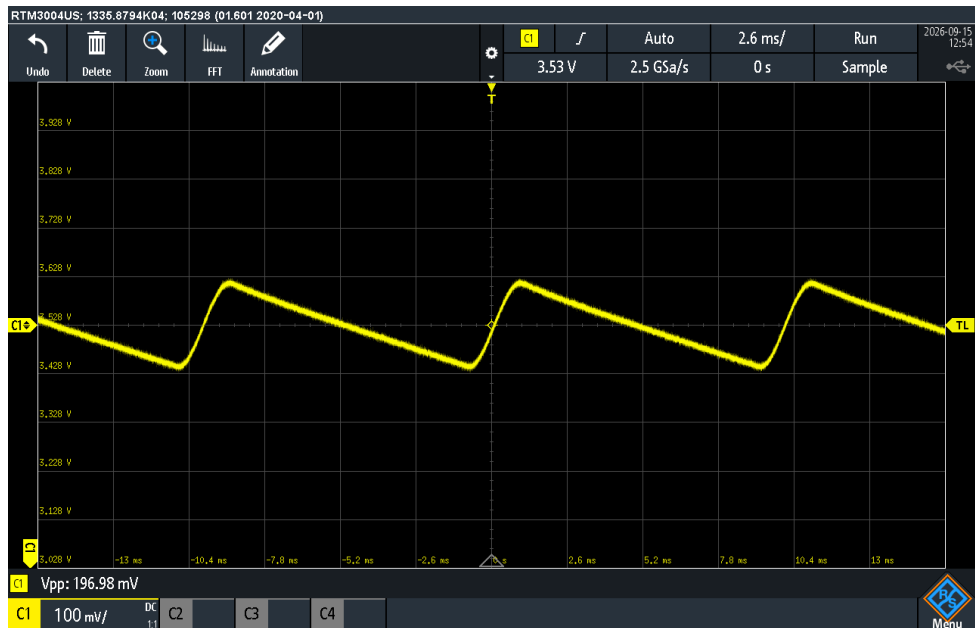


Figure 12: Half-wave rectifier $f = 100 \text{ Hz}$ $A = 5V$

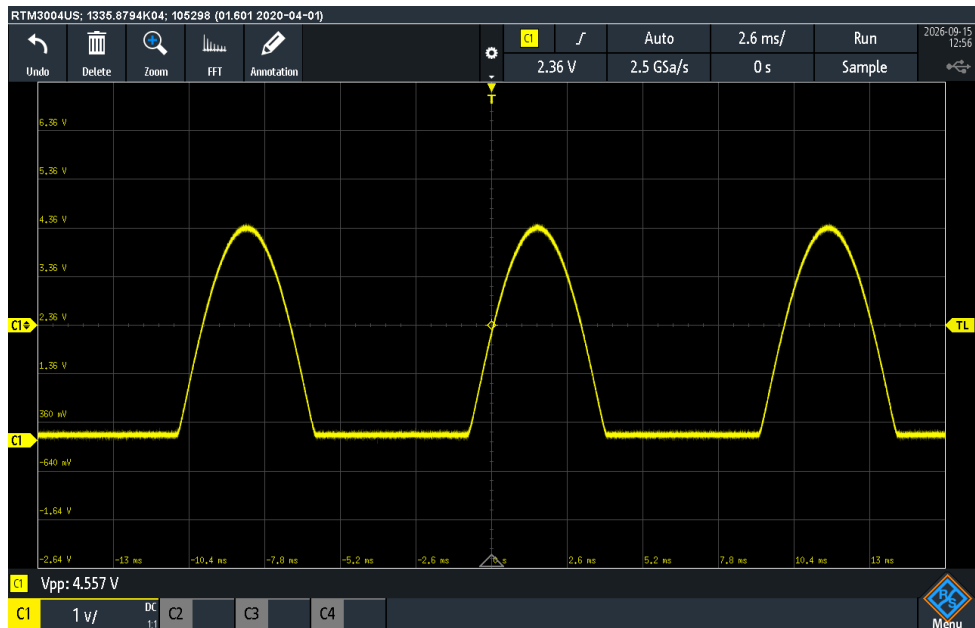


Figure 13: Half-wave rectifier with capacitor removed $f = 100 \text{ Hz}$ $A = 5V$

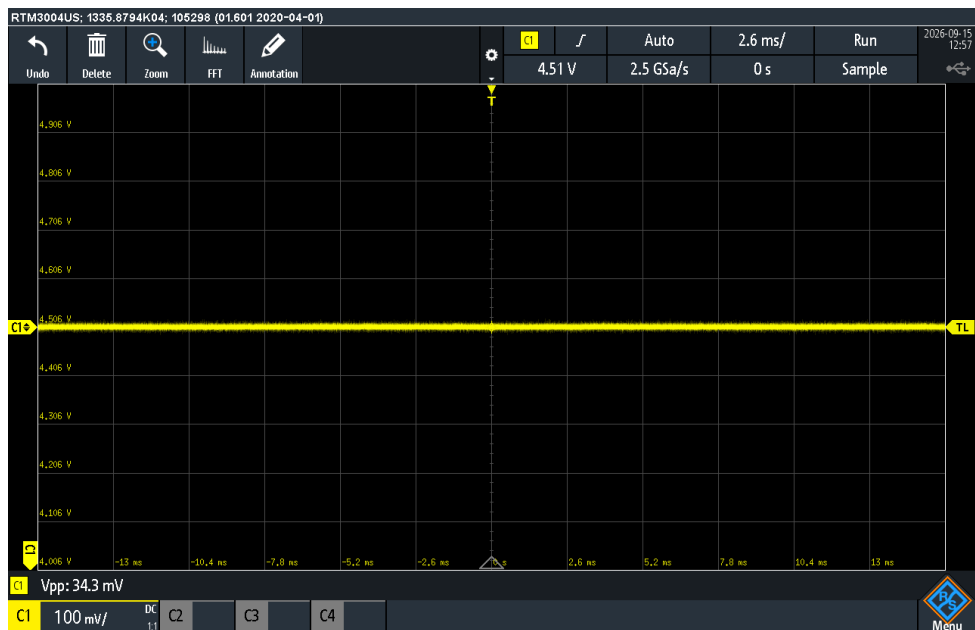


Figure 14: Half-wave rectifier with resistor removed $f = 100 \text{ Hz}$ $A = 5V$

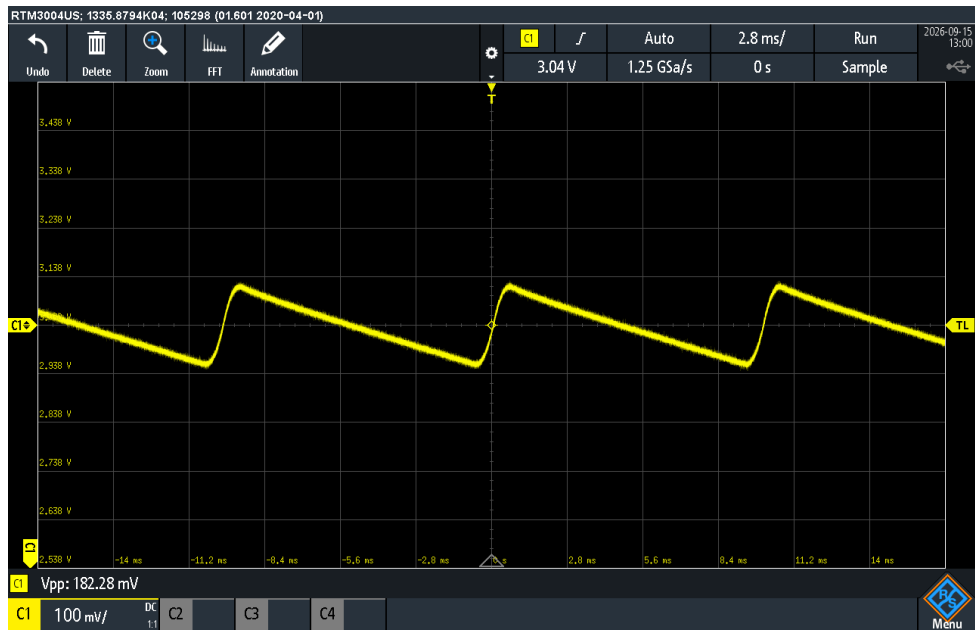


Figure 15: Half-wave rectifier with triangular wave $f = 100 \text{ Hz}$ $A = 5V$

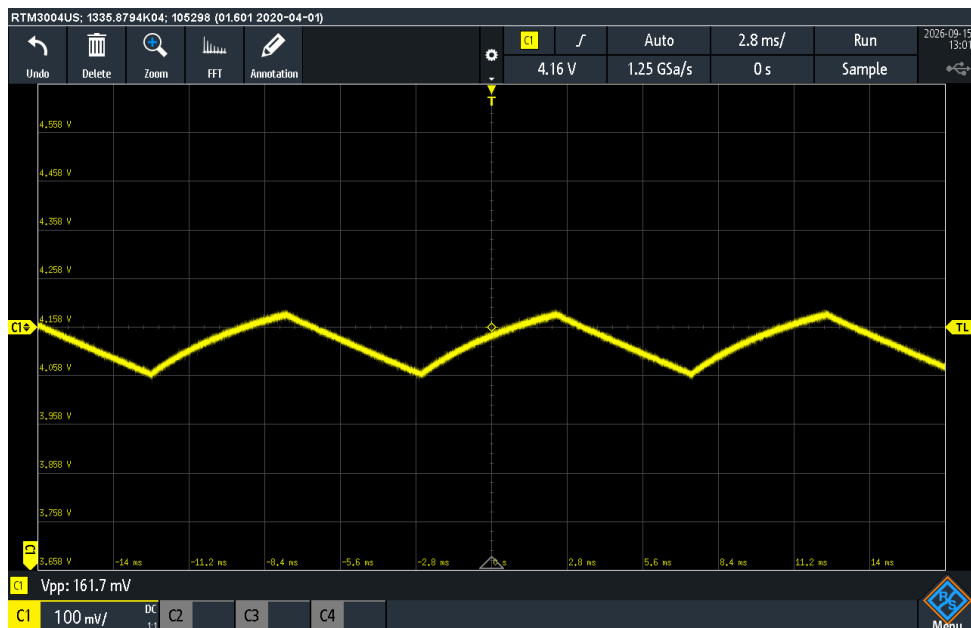


Figure 16: Half-wave rectifier with square wave $f = 100 \text{ Hz}$ $A = 5 \text{ V}$

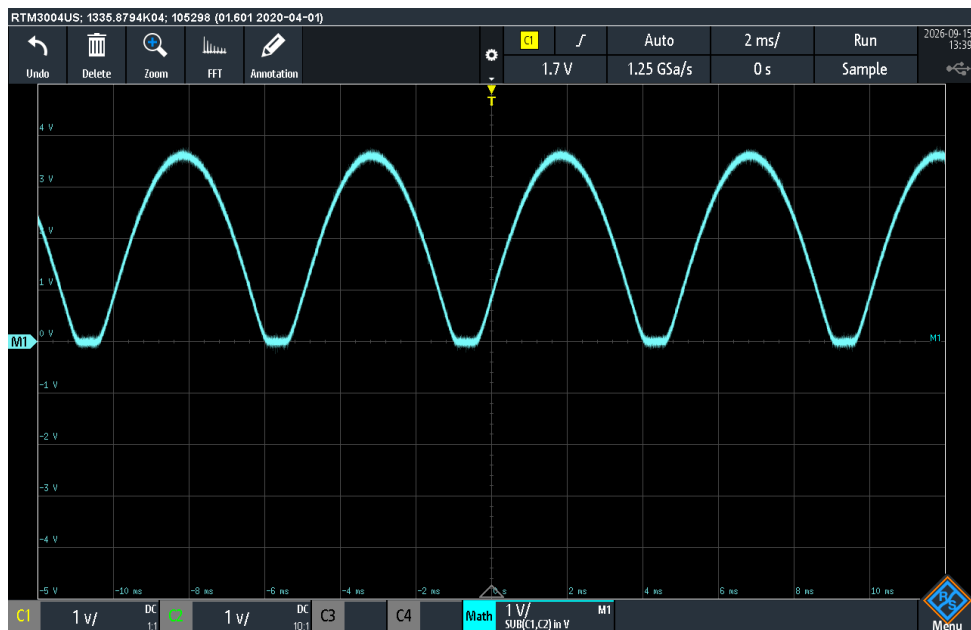


Figure 17: Full-wave rectifier without capacitor $f = 100 \text{ Hz}$ $A = 5 \text{ V}$

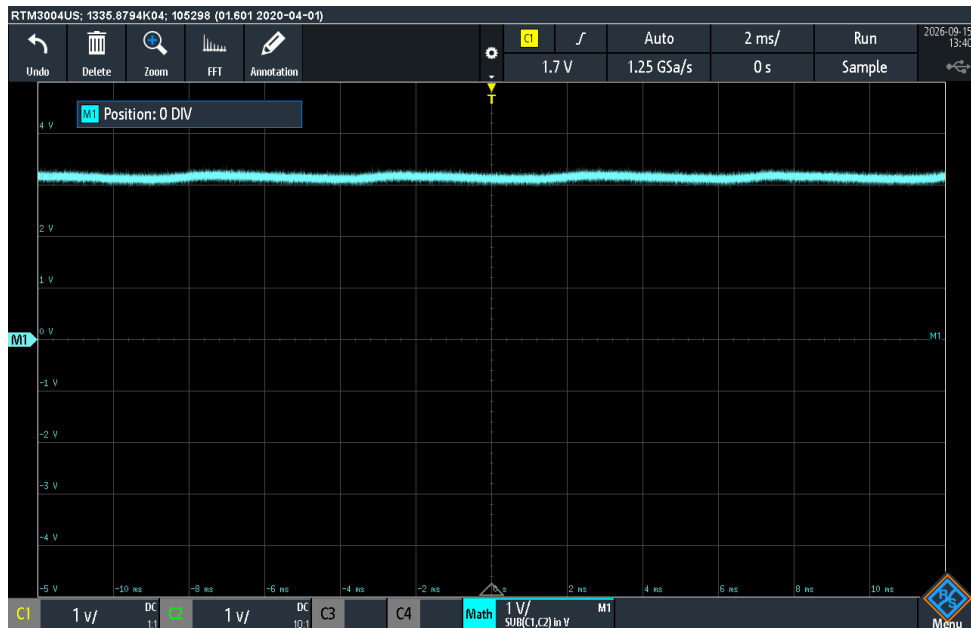


Figure 18: Full-wave rectifier with capacitor $f = 100 \text{ Hz}$ $A = 5V$

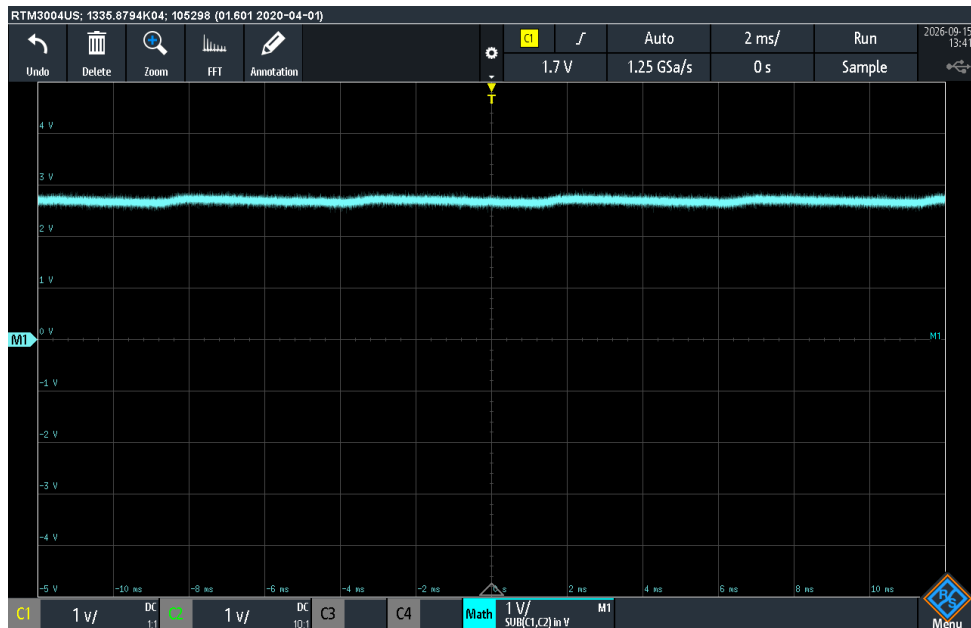


Figure 19: Full-wave rectifier with triangular wave $f = 100 \text{ Hz}$ $A = 5V$



Figure 20: Full-wave rectifier with square wave $f = 100 \text{ Hz}$ $A = 5V$

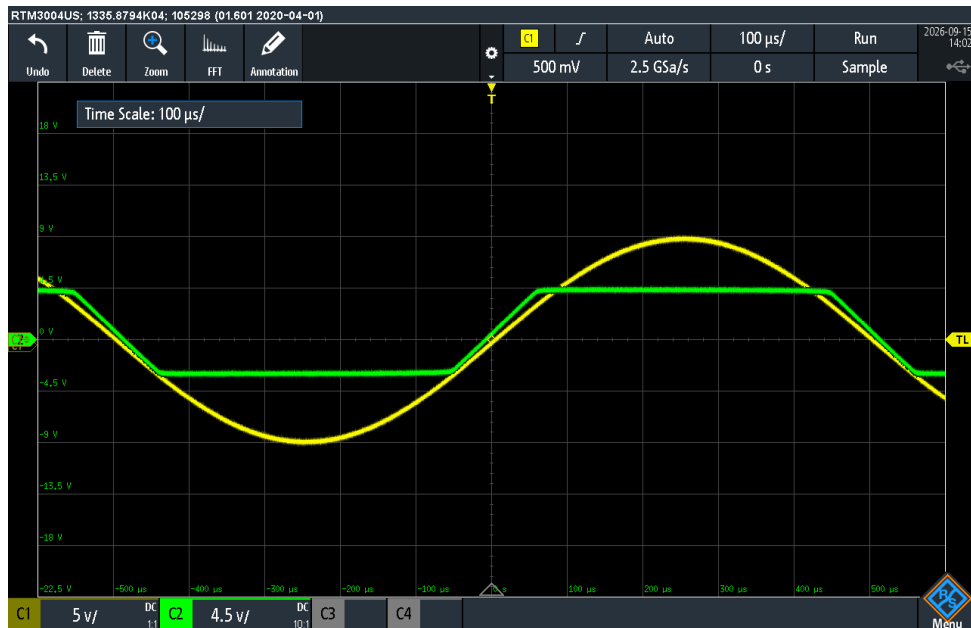


Figure 21: Clipping circuit 1 $f = 1\text{kHz}$ $A = 10V$

Ch1: V_{in}

Ch2: V_{out}



Figure 22: Clipping circuit 1 $f = 1\text{kHz}$ $A = 5\text{V}$

Ch1: V_{in}

Ch2: V_{out}



Figure 23: Clipping circuit 1 $f = 1\text{kHz}$ $A = 2\text{V}$

Ch1: V_{in}

Ch2: V_{out}

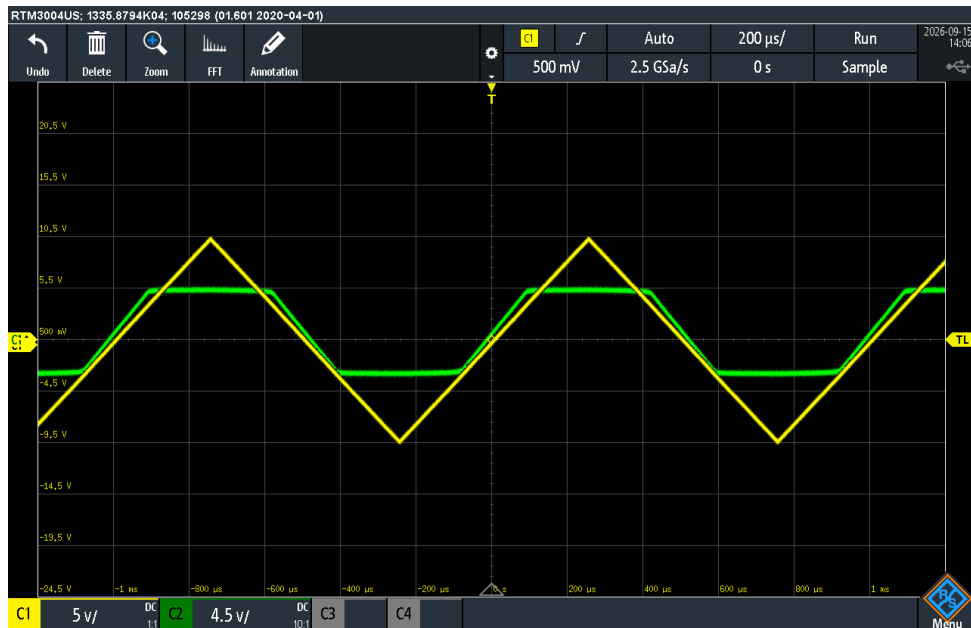


Figure 24: Clipping circuit 1 with triangular wave $f = 1\text{kHz}$ $A = 10\text{V}$

Ch1: V_{in}
Ch2: V_{out}

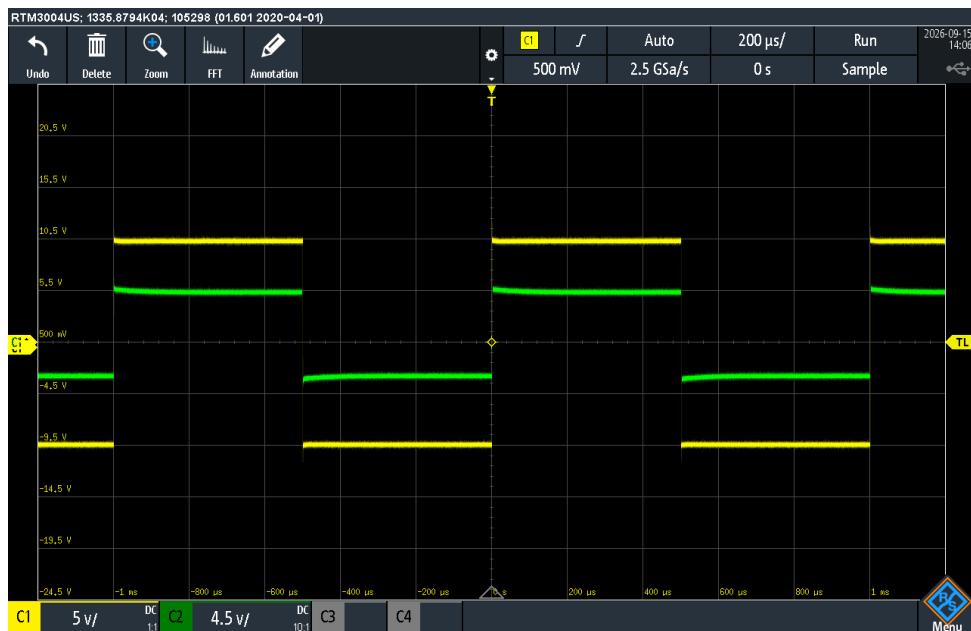


Figure 25: Clipping circuit 1 with square wave $f = 1\text{kHz}$ $A = 10\text{V}$

Ch1: V_{in}
Ch2: V_{out}

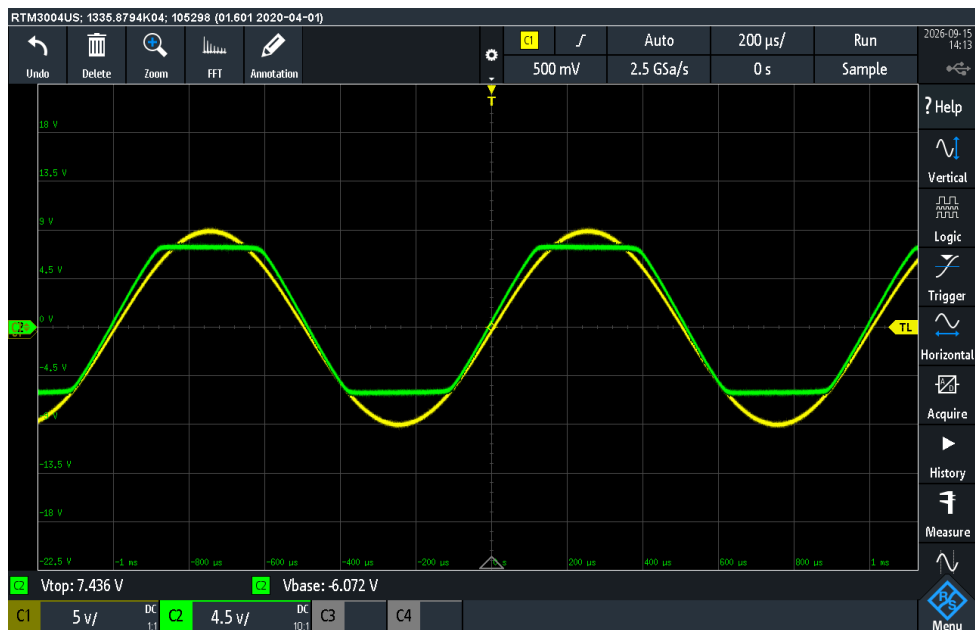


Figure 26:

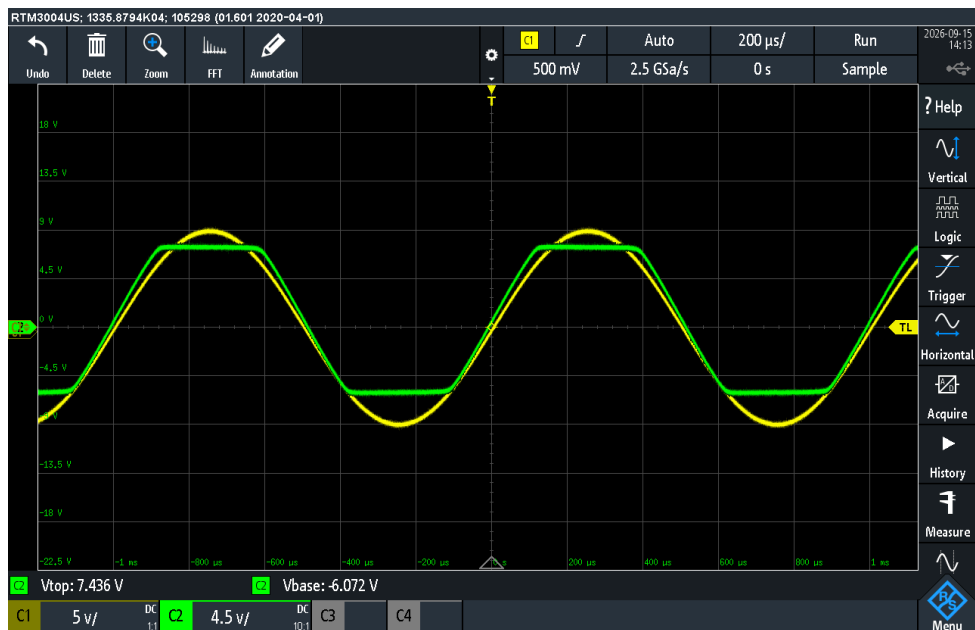


Figure 27:

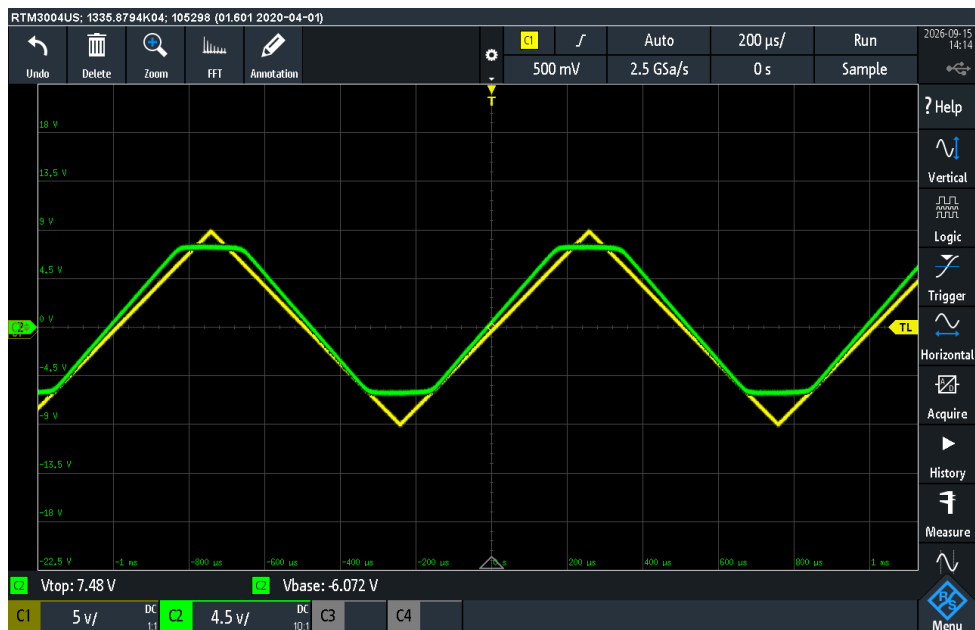


Figure 28:



Figure 29:

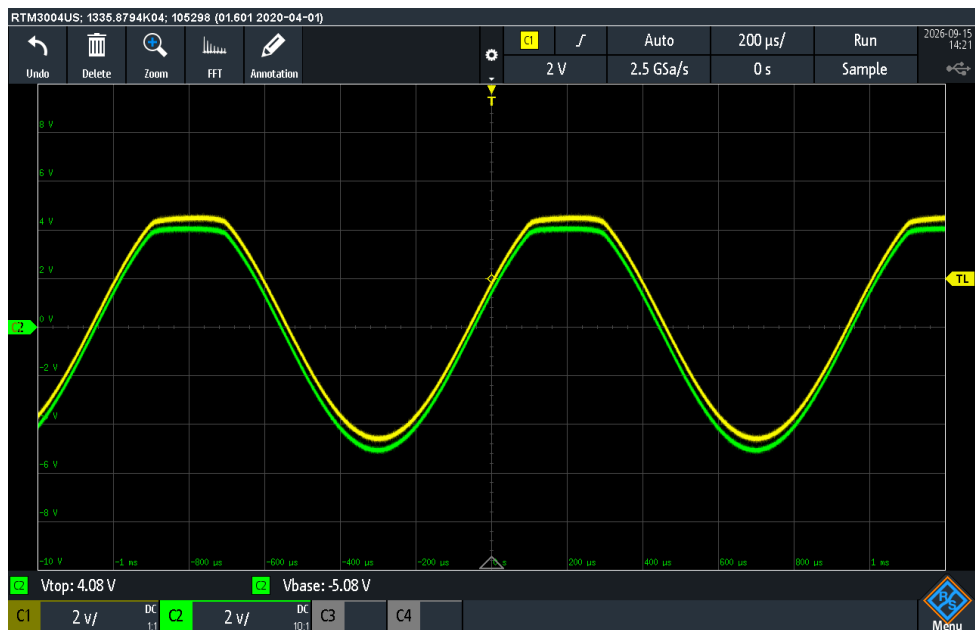


Figure 30:

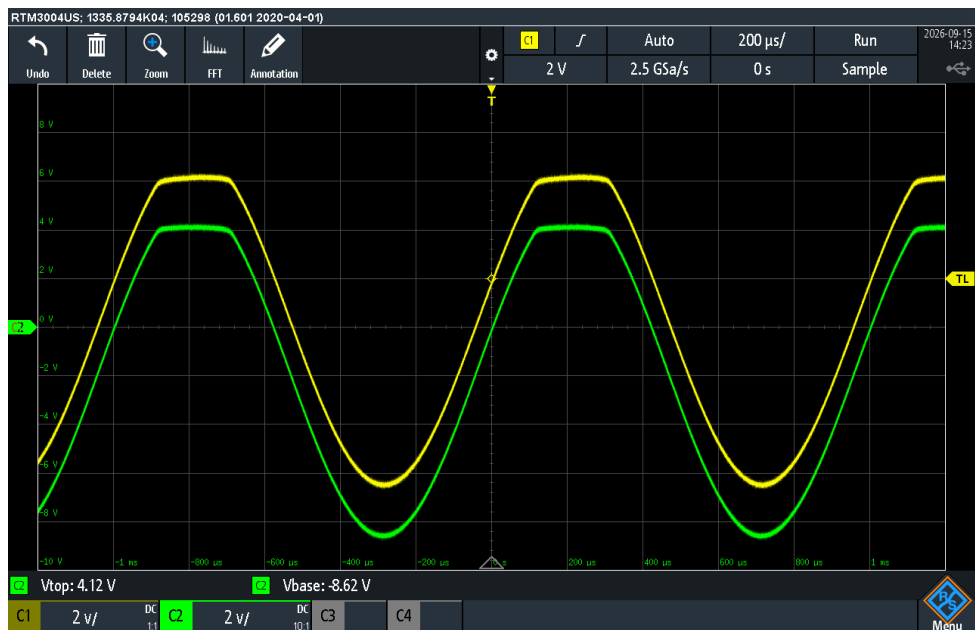


Figure 31:

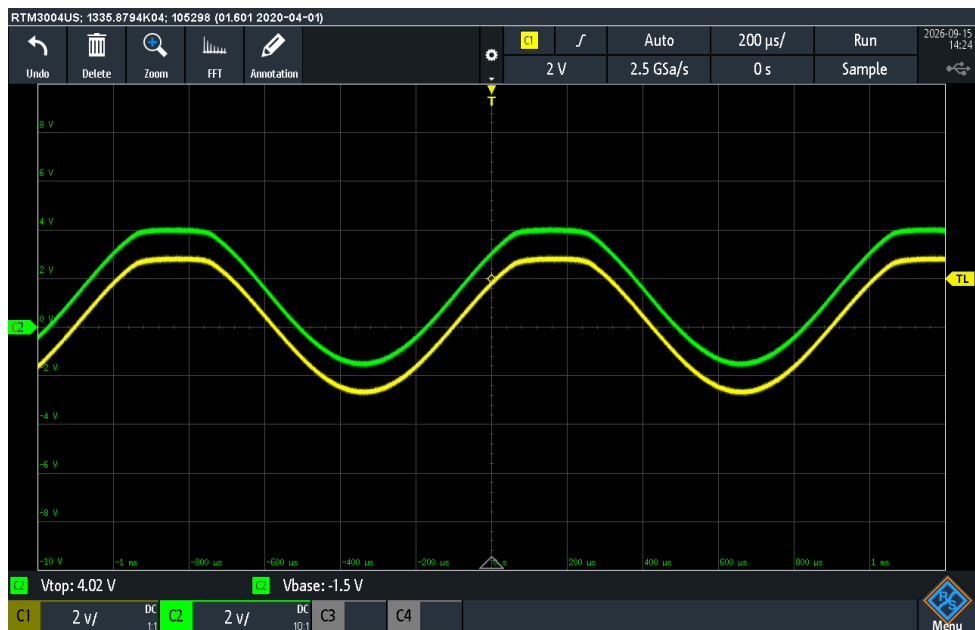


Figure 32:

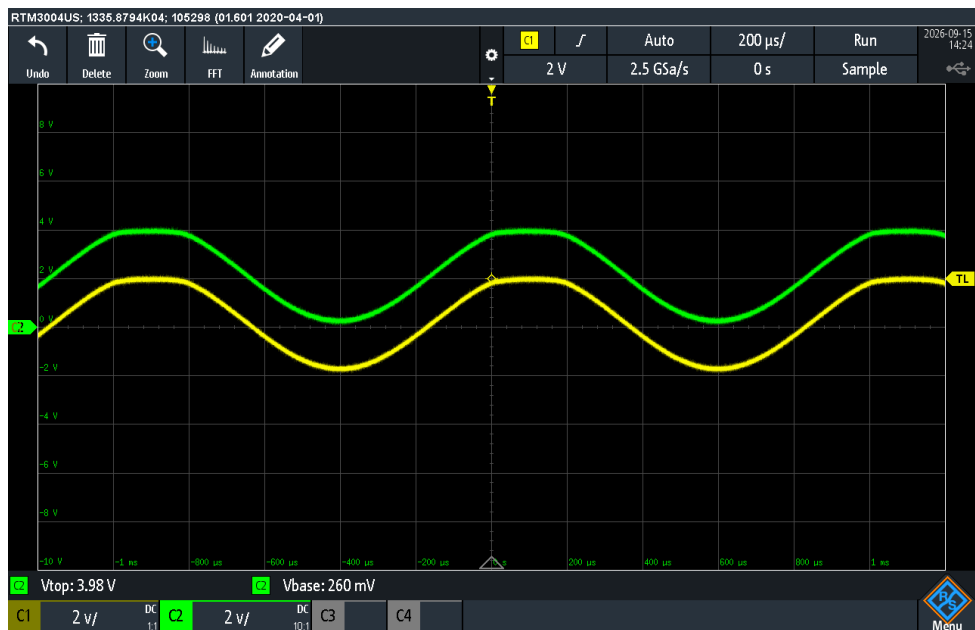


Figure 33:

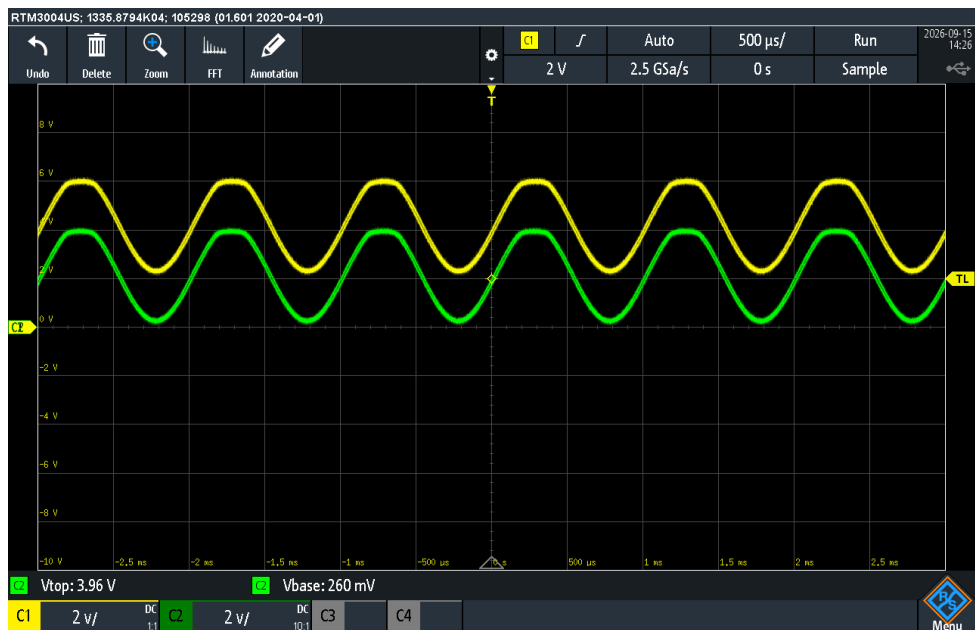


Figure 34:



Figure 35:



Figure 36:



Figure 37:

2.D Conclusion

In this lab, we studied the characteristics and applications of PN junction diodes. We simulated and built half-wave and full-wave rectifiers, clipping circuits, and clamping circuits. The experimental results closely matched the simulation results, confirming our understanding of diode behavior in various circuit configurations.